I. Introduction

Stress is a prevalent and consequential aspect of modern life, impacting individuals both mentally and physically. Stress affects mood, behavior, and health. Short-term stress can be adaptive, but long-term exposure can lead to health issues [1]. Recognizing and understanding stress is crucial for maintaining well-being, and technological advancements have enabled the exploration of innovative approaches to stress detection. This work delves into the domain of stress analysis, focusing on two distinct data modalities: physiological signals and facial expressions.

A. Problem Statement

The challenge addressed in this study is the accurate classification of stress levels using physiological signals and facial expressions. Stress assessment traditionally relies on self-reporting, which can be subjective and prone to biases. Leveraging advanced technologies, such as machine learning models, offers the potential for objective and real-time stress detection. However, developing robust models for stress classification requires overcoming challenges related to diverse data sources, varying time windows, and the intricate nature of stress manifestations.

B. Motivation

The motivation behind this work stems from the potential societal impact of reliable stress detection. By harnessing physiological signals, including electrodermal activity, blood volume pressure, skin temperature, and accelerometer data, alongside facial expressions, we aim to create models that can provide objective insights into an individual's stress state. The outcomes of this research can contribute to the development of proactive stress management strategies, leading to improved mental health and overall well-being.

In this study, we explore a fusion approach aiming to combine the outputs of LSTM and VGG16 models for stress classification using physiological and facial data. Our primary goal is to leverage both data modalities through deep learning models to enhance stress classification, benefiting from their ability to capture distinct patterns. We opt for Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNNs) like VGG16 due to their efficacy in capturing temporal dependencies and image features, respectively. By investigating various models, we aim to thoroughly evaluate their performance in stress classification across different data types and time windows.

II. Literature review

In recent studies, researchers have focused on assessing the emotional response of individuals in various contexts to better understand stress and its implications [2], [3]. For instance, [2] investigates how different car handling setups affect drivers' stress levels, using physiological signals like skin potential responses (SPRs) and electrocardiogram (ECG) signals. Similarly, [3] explores stress alleviation methods using virtual reality (VR) video games, analyzing physiological signals such as ECG, galvanic skin response (GSR), and respiration (RESP) to determine stress levels.

Moreover, there's a growing recognition of the importance of monitoring stress levels among specific demographics, such as college students, due to escalating stress levels in academic and familial settings [4]. This study utilizes physiological signals like photoplethysmography (PPG), ECG, and EEG to assess stress levels among students engaged in Sudoku games under different conditions.

Furthermore, traditional methods of detecting mental stress, such as interviews and questionnaires, may not capture instantaneous emotional responses effectively [5]. Instead, approaches like experience sampling and multi-modal fusion models have shown promise in accurately identifying and classifying stress levels based on physiological signals, facial expressions, and auditory data.

Overall, these studies highlight the significance of understanding stress dynamics and the potential of advanced technologies and methodologies in stress detection and management. Furthermore, our approach draws inspiration from recent studies in stress detection, such as those utilizing multi-modal fusion techniques, where researchers combined different types of data to understand stress levels better. Similarly, we've created a fusion model by combining outputs (probabilities or features) from LSTM and VGG16 models to classify stress. While our work shares the idea of combining data sources for better accuracy, we've tailored our method to suit our specific task, focusing solely on the outputs of these models. This shows how fusion models can be adapted to various research goals and areas.

III. Methodology

The methodology employed for this work includes dataset preprocessing, model implementation and model evaluation. This work attempts to implement the fusion model where two models are combined to classify stress. For physiological data LSTM is used and VGG!6 for facial expressions data.

A. Data Description

Stress analysis from physiological data under pressure: WorkStress3D Dataset [6] is mainly used for this work. For the scope of this study we use only physiological signals data and facial expression data from the WorkStress3D dataset for classifying stress. The physiological signal component file incorporates electrodermal activity (EDA), blood volume pressure (BVP), skin temperature, and accelerometer (ACC) data with the corresponding emotion class (0: stress-free, 1: stressful). There are a total of 3 csv files with these data collected over 15, 30, and 60 seconds data window. Similarly, facial expression data consist of pixels from 48 x 48 grayscale images which are categorized as either stress-free (class 0) or stressful (class 1). Fig. 1 shows the data distribution of the dataset.

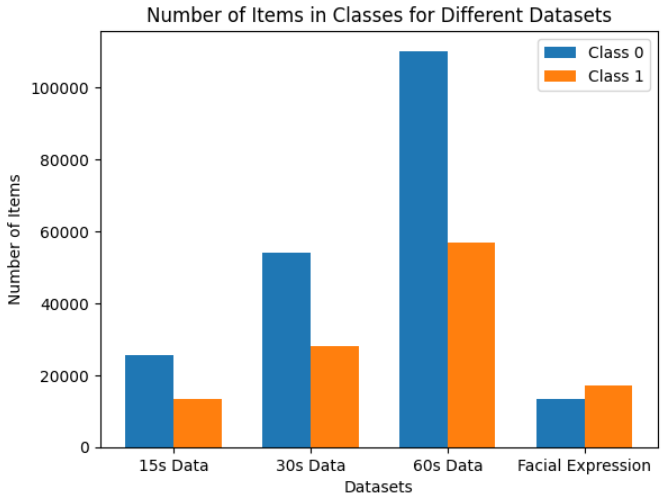


Fig. 1: Graph representing the amount of data

B. Data Preprocessing

The raw data cannot be directly fed to the model so, we have performed some data preprocesing steps to make them compatible for different models.

1) Physiological signals data: The physiological signals were first separated from the emotions to segregate features and labels. To maintain consistency in feature scales, data normalization through standardization using the StandardScaler function from scikit-learn was employed. Finally, data sequences were created for LSTM model.

2) Facial expressions data: After extracting pixel data and converting it into a numpy array, we normalize pixel values to a range between 0 and 1, a standard practice for image data. Emotion labels are encoded numerically and then one-hot encoded to facilitate categorical classification. It is to be noted that the original dataset contains grayscale image pixes so, it must be converted to 3 channels as VGG expects 3 channels.

The dataset is split into three parts: training, fusion, and test sets, each with its own role in developing and testing the models. Table. IV shows how the data is divided.

- Training Set: It consists of physiological and facial data separately to train LSTM and VGG16 models respectively). Additionally, 20% of this dataset is allocated for validation purposes.
- Fusion Set: It is created specifically for training the combined model. It consists of the combined data i.e. both physiological and facial data. This set also includes a validation subset comprising 15% of the data.
- Test Set: It is used for the evaluation of the fusion model's performance. Similar to fusion set, it consists of the combined data.

The number of instances in the fusion and test sets is determined based on the number of facial data instances due to their limited availability. The fusion set, used for training the combined model integrating physiological signals and facial expressions, requires an equal number of data instances from both modalities for effective training. This balance ensures that the fusion model can leverage the complementary nature of these modalities to enhance stress classification accuracy. Similarly, the test set is also fixed in size to maintain consistency with the fusion set, facilitating unbiased evaluation and generalization of the fusion model's performance in classifying stress levels.

IV. Model Implementation

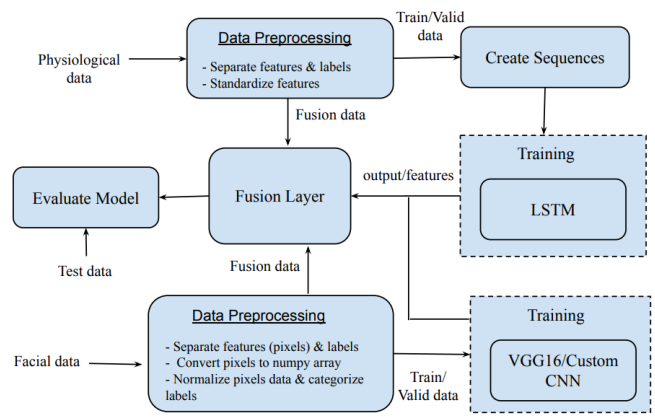


Fig. 2: Procedures for implementing fusion model

In this section, we describe the implementation of the fusion model, which integrates predictions from two individual models trained on physiological signals (LSTM) and facial expressions (VGG16) as shown in Fig. 2. We explore three different approaches for combining the predictions to generate the final stress classification output. The individual models are separately trained at first and then the models are combined and trained on fusion set.

A. Long Short-Term Memory (LSTM) and VGG16

The Long Short-Term Memory (LSTM) model and VGG16 convolutional neural network (CNN) are both integral components of our stress classification system. The LSTM model processes physiological signals data, which are preprocessed and then fed into the model in sequences, as illustrated. Each sequence contains features and their corresponding output labels. The LSTM architecture comprises a single LSTM layer followed by a dense layer, utilizing the sigmoid activation function to predict binary classes. With a total of 18,241 parameters, the model is trained and validated on the preprocessed data, employing the parameters outlined in Table I. Meanwhile, VGG16 is a well-known CNN architecture designed for image classification tasks. In our system, it processes facial expression data. The top layer of the VGG16 model is removed, and additional layers, including global average pooling and dense layers, are added to accommodate our binary classification needs. These modifications enable the model to make predictions using softmax activation. The model parameters used for training and validation are summarized in Table II.

TABLE I: Model Parameters for LSTM

Parameter	Value
Batch size	4
Epochs	30
Memory Units	64
Activation	Sigmoid
Loss function	Binary cross-entropy
Optimizer	Adam
Learning rate	0.001

TABLE II: Model Parameters for VGG16

Parameter	Value
Batch size	128
Epochs	10
Activation	Softmax
Loss function	Categorical cross-entropy
Optimizer	Adam
Learning rate	0.001
Input shape	VGG16(48,48,3)

Both models were fine-tuned based on experimentation to achieve optimal performance in classifying stress levels. The LSTM model operates on physiological signals, while VGG16 handles facial expression data, allowing our system to comprehensively analyze stress from multiple data modalities.

B. Fusion Model

The fusion model is constructed using predictions from the LSTM and VGG16 models, facilitating the integration of physiological signal and facial expression data for stress classification. The fusion layer takes combined predictions from LSTM and VGG16 models as input, followed by dense, and dropout layers and generates classification labels using sigmoid activation. Fig. 3 depicts the architecture of fusion layer. Similarly, Table. shows the model parameters for the

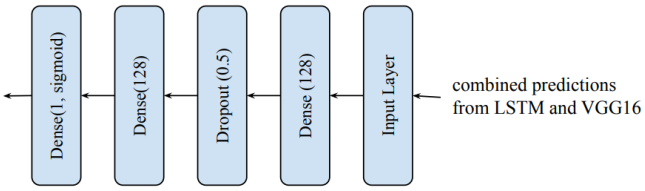


Fig. 3: Fusion layer Architecture

fusion model. These parameters were selected based on experimentation to optimize the fusion model's performance in stress classification.

1) Output Prediction: For the first approach, the fusion model directly utilizes the output predictions generated by the individual LSTM and VGG16 models. These predictions, representing stress classification probabilities, are concatenated and fed into the fusion model. The fusion model then applies a fusion technique, such as dense layers with dropout regularization, to produce the final stress classification output.

TABLE III: Model Parameters for Fusion model

Parameter	Value
Batch size	128
Epochs	10
Activation	Sigmoid
Loss function	Binary cross-entropy
Optimizer	Adam
Learning rate	0.001

2) Features Prediction: In the second approach, the fusion model leverages the intermediate features extracted by the LSTM and VGG16 models. These features capture the representations learned by each model from the input data. For the LSTM model, the fusion model extracts intermediate features using a method that retrieves the output of the second-to-last layer. Similarly, for the VGG16 model, intermediate features are obtained by accessing the output of the third-to-last layer. The fusion model combines these features using dense layers with activation functions, enabling it to capture complex relationships between physiological signals and facial expressions for stress classification.

C. Model Evaluation

F1-score (a combined measure of precision and recall) was used to evaluate the model's ability to accurately classify stress labels. In addition to F1-scores, test accuracy and confusion matrices were generated to delve deeper into the models' performance. Each confusion matrix provides a granular view of the models' strengths and weaknesses, facilitating a comprehensive analysis of their performance in stress classification task. The weighted average score is used to evaluate how well the model performs. It's especially important when there are differences in the number of samples between different classes in the dataset.

V. Experimental Results and Discussion

The models are trained on various combinations of physiological data windows paired with facial data: 15-second data with facial data, 30-second data with facial data, and 60-second data with facial data. Table. IV illustrates the distribution of data across the training, fusion, and test sets utilized in this experiment. As stated in data preprocessing section, the fusion and test sets maintain a fixed size, determined by the quantity of facial data instances, to ensure balanced representation from both modalities. This balance allows for effective training of the fusion model.

TABLE IV: Data Distribution

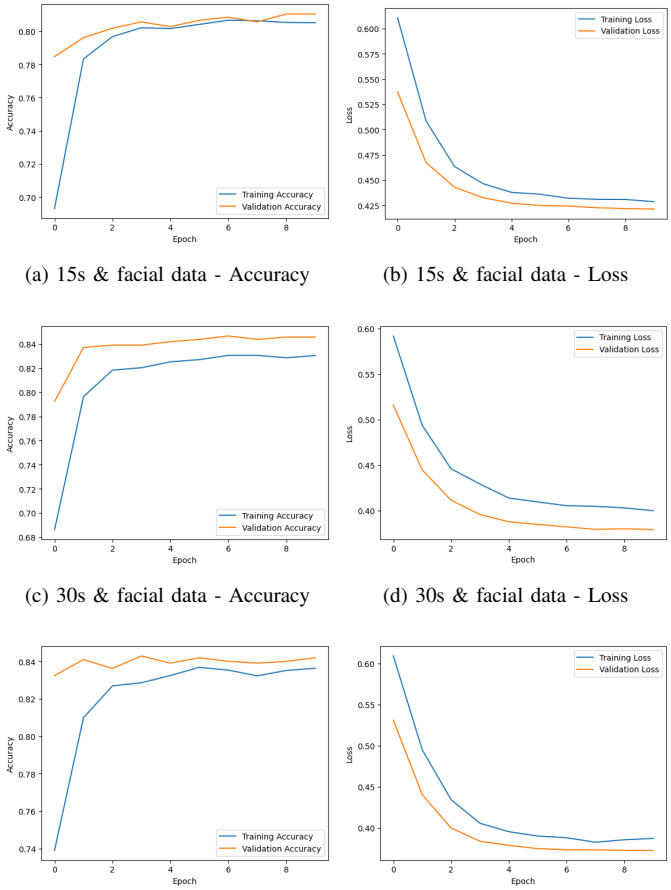
Data	Train	Fusion	Test
Facial (2s)	20,520	7,000	3,000
Physiological (15s)	28,976	7,000	3,000
Physiological (30s)	72,128	7,000	3,000
Physiological (60s)	157,040	7,000	3,000

A. Model Performance

TABLE V: Model Performance on different data combinations

Data	Model	Test acc.	F1-score
15s & facial data	Fusion (output)	0.80	0.80
15s & facial data	Fusion (features)	0.77	0.77
30s & facial data	Fusion (output)	0.82	0.82
30s & facial data	Fusion (features)	0.79	0.79
60s & facial data	Fusion (output)	0.84	0.84
60s & facial data	Fusion (features)	0.78	0.78

The fusion models were evaluated on different combinations of physiological data windows paired with facial data: 15s, 30s, and 60s. Table. V depicts the performance of the fusion models on three windows of physiological data in combination with the facial data. It can be observed that fusion model with output predictions have obtained better scores compared to fusion model with output features. Moreover, Fig. 4 and Fig. 5 shows the plots of the fusion model training and validation which also backs the fusion model with output predictions. Similarly, confusion matrices depicted in Fig. 6 suggests more accurate classification of class-1 labels.



(a) 15s & facial data - Accuracy (b) 15s & facial data - Loss

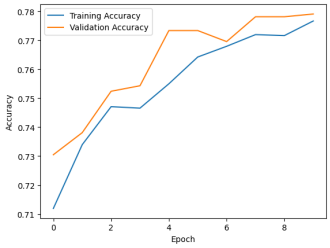
(c) 30s & facial data - Accuracy (d) 30s & facial data - Loss

(e) 60s & facial data - Accuracy (f) 60s & facial data - Loss

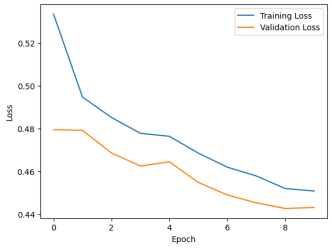
Fig. 4: Training and validation accuracy and loss of Fusion (output) model on different physiological and facial data

B. Findings & Discussion

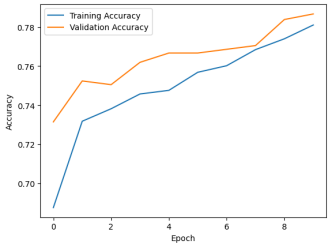
- Across all data combinations, fusion models leveraging output predictions consistently outperformed those with output features in terms of test accuracy and F1-scores. It indicates the efficacy of utilizing raw predictions for stress classification.
- Fusion models with raw predictions consistently outperformed those with extracted features across all data combinations. Notably, for 15s and 30s data, raw prediction models achieved higher accuracy and F1-scores (15s: 0.80/0.77, 30s: 0.82/0.79) compared to feature-based models. Similarly, in the 60s data combination, raw prediction models showed superior performance, yielding an accuracy and F1-score of 0.84/0.78, while feature-based models had slightly lower scores.
- The notable gap in performance between fusion models with output predictions and those with output features suggests that the direct integration of model predictions may capture more relevant information for stress classification than intermediate features.
- The highest performance observed in fusion models using output predictions, particularly evident in the 60s data combination, highlights the potential benefits of leveraging longer physiological data windows for stress classification.
- The confusion matrix also indicates that both the models performed well, especially for class-1 predictions.
- For fusion output predictions, both training and validation accuracies show a steady increase over the training epochs, indicating effective learning. Concurrently, the training and validation loss curves are converging, suggesting successful model training without significant overfitting concerns. In contrast, for fusion output features, the training accuracies exhibit a consistent upward trend, signifying effective learning. However, validation accuracies fluctuate, particularly noticeable with the 60s data combination. Nonetheless, the training and validation loss curves consistently decrease, indicating some degree of convergence.
- These findings underscore the importance of considering



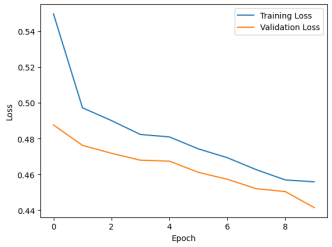
(a) 15s & facial data - Accuracy



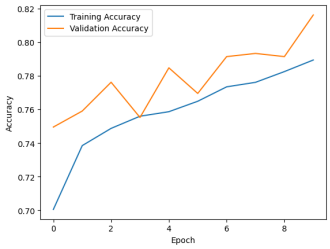
(b) 15s & facial data - Loss



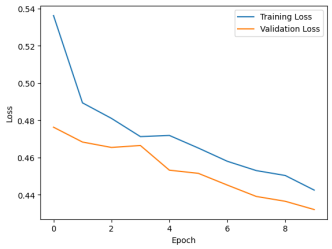
(c) 30s & facial data - Accuracy



(d) 30s & facial data - Loss

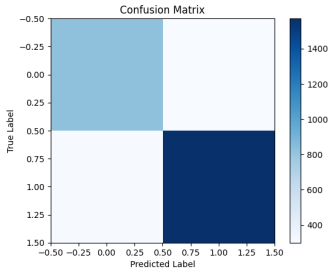


(e) 60s & facial data - Accuracy

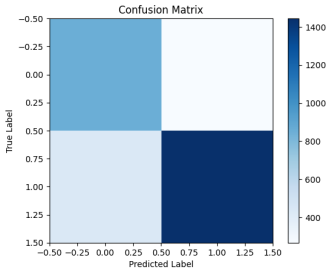


(f) 60s & facial data - Loss

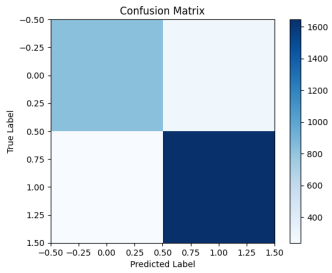
Fig. 5: Training and validation accuracy and loss of Fusion (features) model on different physiological and facial data



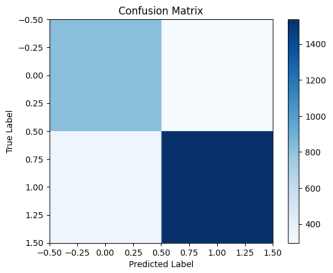
(a) Fusion(output) - 15s & facial data



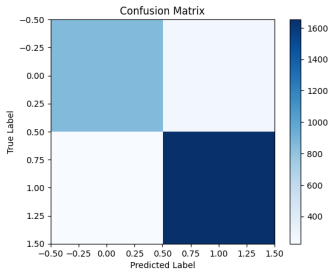
(b) Fusion(features) - 15s & facial data



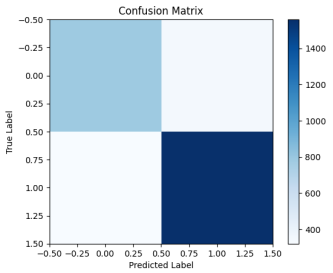
(c) Fusion(output) - 30s & facial data



(d) Fusion(features) - 30s & facial data



(e) Fusion(output) - 60s & facial data



(f) Fusion(features) - 60s & facial data

Fig. 6: Confusion matrices for both fusion models on different physiological & facial data

both the data window length and the type of input data representation when designing fusion models for stress classification tasks.

VI. Conclusion & Future Works

In this study, we explored the effectiveness of a fusion model integrating LSTM and VGG16 architectures for stress classification using physiological signals and facial expressions. Our findings demonstrate that fusion models leveraging raw predictions outperform those using extracted features across various data combinations. Particularly, fusion models with output predictions achieved higher accuracy and F1-scores, indicating the importance of directly integrating model predictions for stress classification. Moreover, our results highlight the potential benefits of leveraging longer physiological data windows, especially evident in the 60s data combination. The fusion model's performance was evaluated comprehensively using test accuracy, F1-score, and confusion matrices, providing insights into its strengths and weaknesses in stress classification. Overall, our study underscores the importance of considering both data window length and input data representation when designing fusion models for stress classification tasks.

Future research could explore the optimization of fusion techniques and architectures to enhance the performance of stress classification models. Additionally, investigating the integration of other modalities such as voice and text data could provide a more comprehensive understanding of stress dynamics. Moreover, incorporating real-time monitoring capabilities and conducting longitudinal studies could further refine the applicability of fusion models in real-world settings.

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